

1. Which of the following forces is NOT electromagnetic in nature?
(a) Force of friction
(b) Tension in a string
(c) Electrical force in spring
(d) Weight of body [d]
2. A bomb explodes on the moon. You will hear its sound on earth after _____
(a) 20 minutes
(b) 2 hrs 20 minutes
(c) 3.7 hours
(d) You cannot hear the sound of explosion [d]
3. A body weighs 700gm weight on the surface of the earth. How much will it weigh on the surface of a planet whose mass is $\frac{1}{7}$ and radius half of the earth
(a) 500 gm
(b) 600 gm
(c) 400 gm
(d) 300 gm [c]
4. If the dimensions of a physical quantity are given by $M^a L^b T^c$, then the physical quantity will be
(a) Force if $a = 0, b = -1, c = -2$
(b) Pressure if $a = 1, b = -1, c = -2$
(c) Velocity if $a = 1, b = 0, c = -1$
(d) Acceleration if $a = 1, b = 1, c = -2$ [b]
5. An ideal Carnot's engine whose efficiency is 40% receives heat at 500 K. If the efficiency is to be 50% then the temperature of the source for the same temperature of the sink will be:
(a) 600 K
(b) 800 K
(c) 1000 K
(d) 1500 K [a]
6. When two small droplets merge together to form a large droplet, then in this process:
(a) Energy absorbed
(b) Energy liberated
(c) Neither liberated nor absorbed
(d) Can't say [b]
7. Consider the effect of temperature on two materials 'P' and 'Q' as, with the rise in temperature, materials 'P' becomes diamagnetic, where 'Q' has no effect of temperature at all, then:
(a) 'P' is paramagnetic and 'Q' is diamagnetic
(b) 'P' is ferromagnetic and 'Q' is paramagnetic
(c) 'P' is paramagnetic and 'Q' is ferromagnetic
(d) 'P' is ferromagnetic and 'Q' is diamagnetic [a]
8. If the potential difference and current through a circuit are doubled, then the resistance of the circuit:
(a) Increases
(b) Goes to zero
(c) Remains the same
(d) Decreases [c]
9. A boy is standing on a rotating table with dumbbells in his hands. He suddenly withdraws his hands to his chest. The angular velocity of the table will
(a) Decrease
(b) Increase
(c) Remain same
(d) Become zero [b]
10. What is the order of wavelength of X-rays, radio waves, micro wave and visible light?
(a) Radio waves < Microwaves > Visible light > X-rays
(b) Radio waves < Microwaves < Visible light < X-rays
(c) Radio waves < Microwaves < Visible light > X-rays
(d) Radio waves > Microwaves > Visible light > X-rays [d]
11. Eight equal drops of water are falling through air with a steady velocity of 10 cm s⁻¹. If the drops combine to form a single drop big in size, then the terminal velocity of this big drop is _____
(a) 40 cm s⁻¹
(b) 10 cm s⁻¹
(c) 30 cm s⁻¹
(d) 80 cm s⁻¹ [d]
12. If the three similar charges of magnitude +Q are placed on the vertices of an equilateral triangle of side x, then the total work done required to arrange the system will be:
(a) $\frac{Q^2}{4\pi \epsilon_0 x}$
(b) $\frac{3Q^2}{4\pi \epsilon_0 x}$
(c) $\frac{2Q^2}{4\pi \epsilon_0 x}$
(d) $\frac{Q^2}{\pi \epsilon_0 x}$ [b]
13. Which among the following region of a transistor is heavily doped?
(a) Junction
(b) Emitter
(c) Collector
(d) Base [b]
14. When a moving bus stops suddenly, the passengers are pushed forward because of the
(a) Friction between the ground and the bus
(b) Friction between the passengers and the ground
(c) Inertia of the passengers
(d) Inertia of the bus [c]

15. The relationship between kinetic energy and linear momentum of an object is
 (a) kinetic energy $\propto$ (linear momentum)⁻¹
 (b) kinetic energy $\propto$ (linear momentum)²
 (c) kinetic energy $\propto \sqrt{\text{linear momentum}}$
 (d) none of the above [b]

16. If the angle of incidence in the first medium is equal to the angle of refraction in the second medium, then the absolute refractive index of the medium is:
 (a) 1.33
 (b) 1.1
 (c) 1.0
 (d) 1.5 [c]

17. A double convex lens has focal length 25 cm. The radius of curvature of one of the surfaces is double of the other. Find the radii if the refractive index of the material of the lens is 1.5.
 (a) 18.75 cm, 37.5 cm
 (b) 50 cm, 100 cm
 (c) 100 cm, 50 cm
 (d) 25 cm, 50 cm [d]

18. Magnetic field due to the current carrying wire as shown in the figure at point "O" will be:



- (a) $\frac{\mu_0 I}{2R}$
 (b) $\frac{\mu_0 I}{4R}$
 (c) $\frac{\mu_0 I}{2\pi R}$
 (d) $\frac{\mu_0 I}{4\pi R}$ [d]
19. The concentric circular coils of 20 turns each are situated in the same plane. Their radii are 25 cm and 45 cm respectively and they carry 0.5 A and 0.9 A current in opposite direction. The magnetic field at the center is:
 (a) Zero
 (b) $\mu_0/20$
 (c) $\mu_0/4$
 (d) $\mu_0/50$ [a]

20. A 100 W light bulb is able to convert 10% of its power to visible radiation. The average intensity of the visible radiation at a distance of 1 m from the bulb is:
 (a) 0.08 W/m²
 (b) 0.8 W/m²

- (c) 10 W
 (d) 8 W/m² [d]

21. Kirchhoff's laws for analysing electrical circuits are based on the conservation of:
 (a) mass and charge
 (b) momentum and energy
 (c) charge and energy
 (d) all of these [c]

22. Which of the following remains constant in an ideal step down transformer?
 (a) Current
 (b) Voltage
 (c) Power
 (d) All of the above [c]

23. The coefficient viscosity of a liquid depends upon :
 (a) both velocity gradient and area of contact
 (b) area of contact
 (c) velocity gradient
 (d) nature of liquid and temperature [d]

24. A gas expands from initial state with P₁ = 320 kPa, V₁ = 0.04 m³ to a final pressure of 80 kPa. If the reversible pressure - volume relation is given by PV² = Constant, then what will be the work done for the process?
 (a) 6.4 kJ
 (b) 64 kJ
 (c) 6.4 MJ
 (d) 0.64 KJ [a]

25. A man standing on a cliff hears the echo of his shout after 1 s. If the velocity of sound in air is 340 m/s, what is the distance between the man and the mountain from which his voice is reflected?
 (a) 340 m
 (b) 170 m
 (c) 680 m
 (d) 85 m [b]

26. If $7 \cos^2 \theta + 3 \sin^2 \theta = 4$ and $0 < \theta < \frac{\pi}{2}$, then what is the value of $\tan \theta$
 (a) $\sqrt{7}$
 (b) $7/3$
 (c) 3
 (d) $\sqrt{3}$ [d]

27. If $y = (\sin x)^y$, then $\frac{dy}{dx}$ is ____:
 (a) $\frac{y^2 \cot x}{1 - y \log(\sin x)}$
 (b) $\frac{y^2 \cot x}{1 - y \log x}$
 (c) $\frac{y^2 \cot x}{1 + y \log(\sin x)}$
 (d) $\frac{y^2 \cot x}{1 + y \log x}$ [d]

28. The centroid of the triangle with vertices (2, 3), (-2, -5) and (3, 5) is at

- (a) (1, 1)
(b) (2, -1)
(c) (1, -1)
(d) (1, 2)

[a]

29. The value of $\int \frac{x^3}{x+1} dx$ is

- (a) $\frac{8}{3} + \ln 3$ (b) $3 + \ln \frac{8}{3}$
(c) $\frac{8}{3} - \ln 3$ (d) $3 - \ln \frac{8}{3}$

[c]

30. The coefficient of the term independent of x in the expansion of $\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^{10}$ is equal to

- (a) 10
(b) 252
(c) 20
(d) 256

[b]

31. Find the area of region bounded by the curve $y^2 = x$ and the line $x = 1$, $x = 4$ and the x-axis

- (a) $\frac{14}{3}$
(b) $\frac{3}{14}$
(c) $\frac{12}{7}$
(d) $\frac{4}{12}$

[a]

32. What is the order and degree of equation

$$\left(\frac{dy}{dx}\right)^{\frac{5}{2}} = \frac{d^3y}{dx^3} \text{ is } \underline{\hspace{2cm}} \text{ respectively.}$$

- (a) 2, 3
(b) 1, 3
(c) 3, 1
(d) 3, 2

[d]

33. Total number of words formed by 2 vowels and 3 consonants taken from 4 vowels and 5 consonants is equal to

- (a) 60
(b) 120
(c) 7200
(d) None of these

[c]

34. Find the probability of getting a sum of 11 when two dice are rolled, given that one of the dice shows a 5.

- (a) $\frac{1}{6}$
(b) $\frac{1}{3}$
(c) $\frac{2}{3}$
(d) $\frac{5}{6}$

[a]

35. If $2A + 3B = \begin{bmatrix} 2 & -1 & 4 \\ 3 & 2 & 5 \end{bmatrix}$ and $A + 2B = \begin{bmatrix} 5 & 0 & 3 \\ 1 & 6 & 2 \end{bmatrix}$,

then B is equal to

- (a) $\begin{bmatrix} 8 & 1 & 2 \\ 1 & 10 & 1 \end{bmatrix}$
(b) $\begin{bmatrix} 8 & 1 & 2 \\ -1 & 10 & -1 \end{bmatrix}$
(c) $\begin{bmatrix} 8 & -1 & 2 \\ -1 & 10 & -1 \end{bmatrix}$

(d) None of these

[b]

36. If $\begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$ then the value of A^{-1} is

- (a) $\begin{bmatrix} i & 0 \\ 0 & -\frac{i}{2} \end{bmatrix}$ (b) $\begin{bmatrix} -\frac{i}{2} & 0 \\ 0 & i \end{bmatrix}$
(c) $\begin{bmatrix} \frac{i}{2} & 0 \\ 0 & -i \end{bmatrix}$ (d) $\begin{bmatrix} i & 0 \\ 0 & \frac{i}{2} \end{bmatrix}$

[b]

37. The 5th term of an AP is 5 and the 13th term is -13. The first term of the AP is

- (a) 11
(b) 12
(c) 13
(d) 14

[d]

38. Foci of ellipse $16x^2 + 144y^2 = 400$ is

- (a) $\pm (3, 0)$
(b) $\pm (4, 0)$
(c) $\pm (0, 3)$
(d) $\pm (0, 4)$

[a]

39. If $\begin{vmatrix} x & 0 & 1 \\ 0 & 2 & 1 \\ 5 & -3 & -1 \end{vmatrix} = 0$ then $x = ?$

- (a) 2
(b) 1
(c) -1
(d) -2

[d]

40. $\int \frac{x}{\sqrt{a^3 + x^2}} dx = ?$

- (a) $\sqrt{a^2 + x^2} + C$
(b) $\frac{1}{\sqrt{a^2 - x^2}} + C$
(c) $\sqrt{a^2 - x^2} + C$
(d) $\frac{1}{\sqrt{a^2 + x^2}} + C$

[]

41. If $\vec{a} = x\hat{j} + y\hat{j} + z\hat{k}$ then $(\vec{a} \cdot \hat{j}) \cdot \hat{j} + (\vec{a} \cdot \hat{k}) \cdot \hat{k} = ?$

- (a) $-\vec{a}$ (b) $\vec{a}$
(c) 1 (d) None of these

[b]

42. If $y = \cos 3x + \sin 5x$ then the value of d^2y/dx^2 is
 (a) $-3 \cos 3x - 5 \sin 5x$
 (b) $-9 \cos 3x - 25 \sin 5x$
 (c) $9 \cos 3x + 5 \sin 5x$
 (d) $9 \cos 3x + 25 \sin 5x$ [b]
43. If $y = ax^2$ then dy/dx at $x = 2$ is equal to
 (a) $4a$
 (b) $-4a$
 (c) a
 (d) $-a$ [d]
44. $\sqrt{\frac{1+\sin A}{1-\sin A}} = \underline{\hspace{2cm}}$.
 (a) $\operatorname{cosec} A - \cot A$
 (b) $\sec A - \tan A$
 (c) $\sec A + \tan A$
 (d) $\sec A$ [c]
45. Determine the distance covered by a particle, given its motion is described by the equation ($s = 490t - 9.8t^2$), until it comes to rest
 (a) 425 unit
 (b) 6125 unit
 (c) 7124 unit
 (d) 638 unit [b]
46. Consider two individuals, A and B, who share the same birthday. What type of relation does this shared characteristic represent between them?
 (a) Reflexive relation
 (b) Symmetric relation
 (c) Transitive relation
 (d) All of the above (Equivalence relation) [d]
47. $\sqrt{-2+2\sqrt{3}i} = ?$
 (a) $\pm (\sqrt{3} + i)$
 (b) $\pm (\sqrt{3} - i)$
 (c) $\pm (1 + i\sqrt{3})$
 (d) $\pm (1 - i\sqrt{3})$ [c]
48. If the point $(2k - 3, k + 1)$ lies on the linear equation $2x - 3y + 12 = 0$ then value of k is :
 (a) 2
 (b) 3
 (c) -3
 (d) -1 [c]
49. If the direction ratios of two lines are $(1, 2, 3)$ and $(-2, 3, -4)$, Then the angle between the lines is:
 (a) $\cos^{-1}\left(\frac{8}{\sqrt{406}}\right)$
 (b) $\cos^{-1}\left(-\frac{8}{\sqrt{406}}\right)$

(c) $\cos^{-1}\left(-\frac{6}{\sqrt{406}}\right)$

(d) $\cos^{-1}\left(\frac{6}{\sqrt{406}}\right)$ [a]

50. What is $\lim_{x \rightarrow 4} \frac{3 - \sqrt{5+x}}{x-4}$ equal to ?

(a) $\frac{-1}{2}$

(b) $\frac{-1}{6}$

(c) $\frac{1}{6}$

(d) $\frac{-1}{5}$ []

51. Select the most appropriate synonym of the given word. **Ambitious**

- (a) aspires
 (b) improve
 (c) decision
 (d) skills [a]

52. Select the most appropriate ANTONYM of the given word. **Exhaust**

- (a) Burn
 (b) Empty
 (c) Collect
 (d) Restore [d]

53. Select the most appropriate synonym of the given word. **GREGARIOUS**

- (a) hostile
 (b) introvert
 (c) sociable
 (d) reserved [c]

54. Choose the correct antonym of the following word from the options given below: **Shallow**

- (a) deep
 (b) long
 (c) short
 (d) brief [a]

55. Choose the most appropriate word that can substitute the given group of words. A person pretending to be somebody he isn't

- (a) Bureaucrat
 (b) Aristocrat
 (c) Conservative
 (d) Hypocrite [d]

56. Select the word which means the same as the group of words given.

One who knows everything

- (a) Omniscient
 (b) Omnipotent
 (c) Omnipatient
 (d) Omnipresent [a]

57. **Select the correctly spelt word.**
 (a) Miscelleneous
 (b) Miscellaneous
 (c) Miscellaneius
 (d) Miscelaneous [b]
58. **Choose an appropriate determiner for the given sentence: I don't have _____ money to spend happily in the tour**
 (a) few
 (b) many
 (c) lot
 (d) little [d]
59. **Fill in the blank with the most appropriate word. We must all be aware _____ the threat to our environment.**
 (a) of
 (b) for
 (c) from
 (d) with [a]
60. **Select the option that expresses the given sentence in indirect speech. He said, "I have got the prize".**
 (a) He said that he has got the prize.
 (b) He said that he had got the prize.
 (c) He said, he have got the prize.
 (d) He said that I had got the prize. [b]
61. **A sentence has been given in Active/Passive Voice. Out of the four alternatives suggested, select the one which best expresses the same sentence in Passive/Active Voice. Who broke the glass pane?**
 (a) The glass pane broken by whom?
 (b) The glass pane is being broken by whom?
 (c) By whom was the glass pane broken?
 (d) By whom was the glass pane being broken? [c]
62. **What is the noun form of the given verb Transform**
 (a) Transform
 (b) Transformation
 (c) Transformed
 (d) Transmit [b]
63. **Select the most appropriate option to substitute the underlined segment in the given sentence. If no substitution is required, select 'No improvement' as your answer. Hardly had he finished then it became bright.**
 (a) there
 (b) when
 (c) if
 (d) No improvement [b]
64. **Find the part of the given sentence that has an error in it. If there is no error, choose 'No error'. There is few information (a)/ that I can (b)/ give you. (c)/ No error (d)**
 (a) (d)
 (b) (a)
 (c) (b)
 (d) (c) [b]
65. **In the following question, some part of the sentence may have errors. Find out which part of the sentence has an error and select the appropriate option. If a sentence is free from error, select 'No Error'. My mother (1)/ has (2)/ grey hairs. (3)/ No Error (4)**
 (a) 1
 (b) 2
 (c) 3
 (d) 4 [c]
66. **Directions- Read the short passage below and answer the questions that follow. The term journalism has its derivative, so to say, it comes from the term journal, which means to register. And hence it means recording of all events. The central point is, it combines writing with news gathering and interpretation. It is true that the journalist's work varies from newspaper to newspaper and from magazine to magazine. All journalists are as much research workers, as they are writers. They cannot write their news or feature stories, unless they locate it and research them first. They must be able to read the in-between lines of the main source news and interpret that. A professionally ethical and successful journalist scans the news for truth by filtering and taking only a small bit of news, say, only 3-5%, may be, coming from an unexpected source. He must be able to see or forecast to himself, the news of tomorrow or the day after, from the news of today because newspapers, falling under yellow journalism, giving sensational news not verifying it, can't sustain its market. Bogus scoops will certainly bring disrepute to the newspaper**
What is journalism?
 (a) record of events
 (b) predict future
 (c) mix past and present
 (d) about writers [a]

67. **Journalists are compared to _____.**
(a) writers of yellow journalism
(b) story writers
(c) writers of anything
(d) research workers [d]
68. **What is the essential point in journalism?**
(a) adding more to attract readers
(b) combining writing with news gathering and interpretation
(c) not combining news writing with gathering and interpreting
(d) unable to read in-between lines [b]
69. **Choose the meaning of the underlined phrase 'yellow journalism'.**
(a) giving sensational news
(b) giving colorful news
(c) giving true account
(d) giving only authorised news [a]
70. **Who is a successful journalist?**
(a) probes partially
(b) mixing false news
(c) one taking 3-5% of news after filtering
(d) write anything that attracts readers [c]